

SOFIA AI ENGINEERING SPECIFICATION

Version 4.0

Phase 1 – AI System Specification

Part IV – Professional Capability Framework

Status: APPROVED (Revision R1)

4.1 Purpose

Defines the Professional Capability Framework, describing the professional work Sofia is engineered to perform and how those capabilities are applied across industries.

4.2 Engineering Philosophy

Sofia is designed around professional work rather than professions. The platform automates repetitive knowledge work while supporting human judgment.

4.3 Professional Capability Model

User Objective → Context Collection → Professional Reasoning → Knowledge Integration → Workflow Construction → Document / Analysis / Decision Support → Human Review → Final Deliverable

4.4 Capability Categories

Executive & Administrative Operations; Business & Strategy; Legal Support; Accounting & Financial Operations; Human Resources; Education; Healthcare Administration; Research & Analysis; AI Engineering; Content Development.

4.5 Intelligent Document Automation

Sofia understands the purpose of a document, gathers required information, validates completeness, applies professional structure, and generates a review-ready draft.

4.6 Workflow Intelligence

Supports workflow engineering by organizing tasks, sequencing activities, identifying dependencies, generating checklists, recommending improvements, and reducing operational complexity.

4.7 Capability Expansion Framework

New professional domains inherit Sofia's Cognitive Architecture, Engineering Principles, Professional Reasoning Framework, and Human-Centered Design Philosophy.

4.8 Design Constraints

Sofia supports rather than replaces qualified professionals and encourages human review for significant decisions while operating within legal, ethical, and organizational requirements.

4.9 Future Professional Domains

Aviation, Manufacturing, Supply Chain, Government, Banking, Insurance, Architecture, Engineering, Pharmaceutical Operations, Scientific Research, Non-Profit Organizations, and International Development.

4.10 Engineering Summary

Sofia transforms structured cognitive reasoning into practical professional assistance by automating repetitive knowledge work and supporting decision-making across industries.

Approved by Project Owner: Ana Gemma (Creator of Sofia AI)

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